

BVC24 ERP

AI SMART MANUFACTURING

Employee Module Documentation

Deployment & User Guide

Bharath Vending Corporation

Vendor-based Manufacturing ERP

Version: 1.0

Date: 6 June 2026

Audience: HR | IT | Employees | Implementers

BVC24 - Employee Module

Deployment-grade reference for the employee portal, attendance, leave, memos, documents, performance, and the floating ERP assistant.

*Audience: HR managers, system administrators, IT operations, support staff, and developers extending the module. Companion document: docs/modules/02-hr.md covers the admin/HR side of the same data. This document focuses on the **employee-facing experience** and **production deployment**.*

Table of Contents

1. Module Overview
2. Architecture & Code Map
3. Roles & Access Matrix
4. Lifecycle: From Onboarding to Daily Use
5. Authentication & Session
6. Employee Portal - Section-by-Section
7. Attendance & Geofencing
8. Leave & Permissions
9. Tasks & Productivity
10. Memos & Acknowledgement
11. Employee Documents
12. Performance, Productivity & Rewards
13. The Floating ERP Assistant (Chatbot)
14. Employee API Reference
15. Notifications & Email
16. Production Deployment Checklist
17. Security Considerations
18. Troubleshooting Runbook
19. HR / Admin Operational Tasks
20. Extending the Module
21. Appendix A - localStorage Keys
22. Appendix B - State Transitions
23. Appendix C - Quick Reference for Employees

1. Module Overview

The Employee Module is the part of BVC24 that every shop-floor worker, supervisor and HR officer touches every day. It bundles:

Sub-module	Employee can	HR can
Login	Sign in with EMPLOYEE_CODE + password	Create accounts, reset passwords
Profile	Complete one-time self-registration form, upload photo	Edit any field, view full profile

Sub-module	Employee can	HR can
Attendance	Check in / check out (web or biometric kiosk) with geofence validation	View full ledger, mark absent, audit GPS failures
Tasks	Accept/reject assignments, mark progress (Pending -> In Progress -> Completed), put on hold	Assign tasks (manual or auto), track workload, view performance
Leave	Apply for day-leave or hourly permission, view balance, cancel pending requests	Approve via emailed link (1-click), reject with reason, view all requests
Memos	Read & acknowledge memos (warnings, appreciations, disciplinary notices)	Issue memos, attach files, close/cancel, export to CSV
Documents	View own documents (read-only)	Upload identity proofs, certificates, offer letters
Performance	See own score, productivity chart, ratings, rewards (points, streak, badge)	See company-wide performance, suggested increments
Chatbot	Multi-turn leave application, check status, ask any ERP question	Same - plus HR-wide overviews (pending approvals, on-leave-today)

Design principle: the portal is **read-mostly + structured-write**. Employees never directly edit core records (department, designation, salary, role). They request changes; HR approves.

2. Architecture & Code Map

2.1 Frontend (React 19 + Vite)

```

frontend/src/
|--- App.jsx                # Routing + ProtectedRoute + RoleBasedLanding
|--- pages/
|   |--- Login.jsx          # Admin / Employee tabbed login
|   |--- EmployeeDashboard.jsx # Main portal (role="employee" landing)
|   |--- EmployeeProfileForm.jsx # One-shot self-registration gate
|   |--- EmployeeMemos.jsx   # Memo viewer (scoped to current employee)
|   |--- EmployeeOnboardingChat.jsx # Public token-gated onboarding flow
|   |--- ApplyLeave.jsx       # Standalone leave application page
|   |--- BiometricCheckIn.jsx # Kiosk page (public, no auth)
|   L-- Attendance.jsx      # Read-mostly attendance view
|--- components/
|   |--- ChatBot.jsx         # Floating ERP Assistant (SSE streaming)
|   L-- GeofenceGate.jsx     # GPS-based attendance gate
L-- services/
    L-- api.js               # Axios instance + same-host autodiscovery

```

2.2 Backend (FastAPI + SQLAlchemy + MySQL)

```

backend/app/routes/
|--- connect.py          # /admin-login, /employee-login, /employee-logout
|--- employee.py         # CRUD + photo + reset-password
|--- employee_portal.py  # /employee/{id}/portal-dashboard (one-shot)
|--- employee_task.py    # task accept/reject/status, today-task, all-tasks
|--- employee_documents.py # documents CRUD
|--- employee_memos.py   # memos CRUD + acknowledge + CSV export
|--- employee_onboarding.py # token-gated public onboarding chat
|--- attendance.py       # /check-in, /check-out, /attendance/*
|--- leave.py            # /leave/apply, /balance, /my-requests, /apply-permission,
/cancel, /decide
|--- geofence.py         # /geofence/settings, /validate, /log-failure,
/security-logs
L-- chatbot.py           # /chat/stream (SSE) + leave bridge

backend/app/services/
|--- leave_chatbot_service.py # Multi-turn leave NLU + slot filling
|--- payroll_service.py      # Monthly slip generation
L-- performance_service.py   # Star rating + increment %

backend/app/models/models.py # SQLAlchemy ORM definitions

```

2.3 Database Tables

Table	Anchored to	Notes
employee	-	Master record. PROFILE_SUBMITTED gates the self-reg form
attendance	employee	One row per employee per day. CHECK_IN, CHECK_OUT, STATUS, GEOFENCE_*
leave_request	employee	All leaves + permissions. STATUS ∈ {PENDING_APPROVAL, APPROVED, REJECTED, CANCELLED}
leave_balance	employee	Yearly CASUAL / SICK / EARNED / MATERNITY quotas
task_assignment	employee + project	Assigned work. TASK_STATUS, APPROVAL_STATUS
employee_memo	employee	Warnings, appreciations, disciplinary. Soft-deletable
employee_document	employee	Identity, education, offer letter - file references
biometric_event	employee (via FINGERPRINT_ID)	Raw scans, audit log
notification	employee	In-app inbox
geofence_settings	vendor	Per-vendor office centre + radius
attendance_security_log	employee (nullable)	Failed GPS / geofence attempts

3. Roles & Access Matrix

The system has **two top-level roles** stored in `localStorage.role`: "admin" and "employee". The landing route / dispatches via `App.jsx:44 -> RoleBasedLanding()`.

Action	Employee	HR / Admin
View own dashboard	[OK]	- (admins see Dashboard.jsx)
Edit own profile (one-shot self-reg)	[OK]	[OK] (anytime via Employees page)
Change own department / role / salary	[X]	[OK]
Check in / check out	[OK]	[OK] (any employee)
View own attendance	[OK]	[OK] (all)
Apply for leave / permission	[OK]	[OK] (on behalf of any employee)
Cancel own pending leave	[OK]	[OK]
Approve / reject leave	[X]	[OK] (via emailed token link, 1-click)
Accept / reject task assignment	[OK]	-
Mark task IN_PROGRESS / COMPLETED / ON_HOLD	[OK] (own tasks)	[OK] (any)
Issue / edit / close memos	[X]	[OK]
Acknowledge memo	[OK] (own only)	[OK]
Upload own documents	[X]	[OK] (uploads on employee's behalf)
View own documents	[OK]	[OK]
View own performance	[OK]	[OK] (all)
View company performance	[X]	[OK]
Use floating chatbot	[OK]	[OK]
Multi-turn leave via chatbot	[OK] (auto-fills self)	[OK]
Use HR-wide chatbot queries	[OK] (read-only)	[OK]

Important: Most backend endpoints do **not** enforce JWT yet - the access matrix is largely **frontend-driven**. The login flow stamps `role` into `localStorage`, and the UI only renders employee-appropriate controls. See §17 Security Considerations for hardening recommendations before public deployment.

4. Lifecycle: From Onboarding to Daily Use

4.1 Account Creation (HR action - once per employee)

```

HR opens    Dashboard -> Employees -> Add Employee
HR fills    EMPLOYEE_CODE, NAME, EMAIL, PHONE, PASSWORD,
            DEPARTMENT, DESIGNATION, ROLE, JOINING_DATE,
            SALARY, SHIFT_START / END
Backend     POST /create-employee
            - bcrypt-hashes PASSWORD
            - sets PROFILE_SUBMITTED = 0
HR shares   EMPLOYEE_CODE + initial password with the employee
            (in-person, SMS, or secure channel)

```

4.2 First Login (Employee action - once)

```

1. Employee opens https://<your-host>:5173/login
2. Selects "Employee" tab -> enters EMPLOYEE_CODE + password
3. Backend POST /employee-login validates, returns:
   - JWT token
   - EMPLOYEE_ID, EMPLOYEE_NAME, DEPARTMENT
   - ATTENDANCE_STATUS, HAS_PENDING_FROM_YESTERDAY
   - Side-effect: creates today's Attendance row (CHECK_IN)
4. Frontend stamps localStorage:
   auth=true, role=employee, token=<jwt>,
   employee_id=<UUID>, employee_name=...
5. / route loads EmployeeDashboard.jsx
6. Dashboard hits GET /employees/by-code/{code}
   - If PROFILE_SUBMITTED=0 -> renders EmployeeProfileForm
   - If PROFILE_SUBMITTED=1 -> renders full dashboard
7. Employee fills 7-section form, optionally uploads photo
8. POST /employees/by-code/{code}/submit-profile
   - Flips PROFILE_SUBMITTED = 1
   - Admin-gated fields (ROLE_ID, SALARY) silently dropped
9. Dashboard re-renders in normal mode

```

4.3 Daily Use

```

Morning
Login -> automatic check-in stamped
Geofence validated (if enforcement ACTIVE) - see §7
System computes STATUS: PRESENT / LATE
If LATE: auto-creates LATE_COMING Permission row

Throughout day
View "Today's Tasks" -> click Start Task -> IN_PROGRESS
Complete task -> award points, unlock next stage
Apply for leave via inline form OR chatbot
Read & acknowledge memos
Ask chatbot for status, balances, anything

Evening
Click "Logout" or close tab
POST /employee-logout stamps CHECK_OUT
Computes WORKED_HOURS + OVERTIME_HOURS
If left early: auto-creates EARLY_EXIT Permission

```

4.4 Self-Onboarding Variant (Token-Gated)

For new hires who haven't been physically onboarded yet:

```

HR creates onboarding token (admin flow generates URL)
HR emails: https://<host>:5173/employee-onboarding/<token>
Employee opens link (public, no login)

```

EmployeeOnboardingChat collects: name, email, phone, address,
education, experience, skills
On submit -> creates employee record with default password
Employee then proceeds to normal login flow

5. Authentication & Session

5.1 Login Endpoints

Endpoint	Used by	Auth	Response
POST /admin-login	Admin tab	EMAIL or USERNAME + PASSWORD	token, role, permissions
POST /employee-login	Employee tab	EMPLOYEE_CODE or UUID + PASSWORD	token, EMPLOYEE_ID, name, department, attendance_status
POST /employee-onboarding/{token}/login	Onboarding flow	token + DOB or PHONE	scoped session
POST /employee-logout	Employee	JWT (current session)	check-out stamped

5.2 Session Storage

After successful login, the frontend stamps `localStorage`:

Key	Set by	Read by
auth	Login	ProtectedRoute
role	Login	RoleBasedLanding, sidebar filter
token	Login	Axios request interceptor (Bearer)
employee_id	Employee login	Dashboard, ChatBot, ApplyLeave
employee_name	Employee login	Dashboard greeting, profile card
department	Employee login	Profile card
employee_role	Employee login	Profile card (e.g. "Operator", "Supervisor")
loginTime	Employee login	Session age display
attendance_status	Employee login	"Already checked in" indicator
pending_yesterday	Employee login	Carry-over task banner
chatbot_leave_state_{employee_id}	ChatBot leave flow	Per-employee multi-turn state

5.3 401 Handling

`API.interceptors.response` in `api.js:54` intercepts 401: - `localStorage.clear()` - if role was employee
-> redirect to /login

5.4 Backend Host Auto-Discovery

api.js:16 auto-detects the backend:

1. VITE_API_URL env var (production override)
2. Same-host autodetection: `${proto}://${window.location.hostname}:8001`
3. Hardcoded `http://127.0.0.1:8001` fallback

This means a phone on `192.168.1.20` hitting `http://192.168.1.56:5174` automatically targets `http://192.168.1.56:8001` - no code change needed.

6. Employee Portal - Section-by-Section

The portal at `EmployeeDashboard.jsx` is one continuous scrolling page. Sections (top to bottom):

6.1 Profile Card

Photo (or initials avatar), name, `EMPLOYEE_CODE`, performance rating (*5), productivity badge. Click photo to upload.

6.2 KPI Grid (8 tiles)

Total Tasks - Today - Pending - In Progress - On Hold - Completed - Upcoming - Overdue

6.3 Today's Tasks (sticky)

Shows tasks due today. Action buttons: - **Start Task** -> status -> `IN_PROGRESS`, stamps `START_TIME` - **Complete** -> status -> `COMPLETED`, stamps `END_TIME`, awards points, unlocks next stage - **Hold** -> status -> `ON_HOLD`

6.4 Tabbed Task List

Pending - In Progress - On Hold - Upcoming - Completed (each tab shows count).

6.5 Assigned Projects Grid

Cards per project with progress bars per stage (from `workOrderStageProgress`).

6.6 Performance Breakdown

8 metrics with horizontal bars: - On-time delivery %, Quality score, Attendance %, Task completion rate, Average completion time, Stage-flow efficiency, Memo-impact score, Final composite

6.7 Productivity Chart

6-month line chart of tasks completed per month (Recharts).

6.8 Attendance Summary

4 tiles: Present - Absent - On Leave - Permissions Used - Monthly % bar.

6.9 Rewards

Points balance, current streak (consecutive days punctual), badge (Bronze/Silver/Gold/Platinum).

6.10 My Memos

List of memos issued to the employee. Each unacknowledged memo has an "Acknowledge" button -> modal with optional remark -> `POST /memos/{id}/acknowledge`.

6.11 Leave & Permission (inline forms)

Quick-apply forms below the dashboard, also accessible via chatbot.

6.12 Floating Chatbot

Bottom-right red bubble. Click to open the ERP Assistant panel - see §13.

7. Attendance & Geofencing

7.1 Three Check-In Channels

Channel	Path	How
Web login	POST /employee-login	Automatic on successful login (creates Attendance row)
Web check-in	POST /check-in	Manual button on Attendance page (with GPS)
Biometric kiosk	POST /biometric/scan	Tablet at entrance + fingerprint reader

7.2 Geofence Enforcement

GeofenceSettings table holds the office centre lat/lng + radius. Flow:

1. Page mounts -> GeofenceGate calls GET /geofence/settings
2. If IS_ACTIVE=false -> bypass; allow check-in from anywhere
3. If IS_ACTIVE=true -> browser navigator.geolocation.getCurrentPosition()
 - Success -> POST /geofence/validate with lat/lng
 - > if allowed: enable check-in button
 - > if not : show "Outside Office Geofence (3.2km)"
 - Denied / unavailable / timeout
 - > show error + "Skip GPS" button
 - > POST /geofence/log-failure (security audit)
 - > Status logged as GEOFENCE_STATUS=UNKNOWN

7.3 What Gets Stamped

Every successful check-in fills these attendance columns: - CHECK_IN, STATUS (PRESENT / LATE), LATITUDE, LONGITUDE, DISTANCE_METERS, GEOFENCE_STATUS, DEVICE_INFO, BROWSER_INFO

Every check-out fills: - CHECK_OUT, WORKED_HOURS, OVERTIME_HOURS, CHECKOUT_DISTANCE_METERS

7.4 LATE / EARLY_EXIT Auto-Permissions

If check-in is past SHIFT_START + grace, the system silently creates a LEAVE_TYPE=PERMISSION, SUBTYPE=LATE_COMING row for HR review. Same for early checkout -> EARLY_EXIT. No quota deduction; just an audit trail and a notification.

7.5 Mobile Testing Setup

For testing geofence from a real phone on the same Wi-Fi (very common deploy scenario):

1. **Backend** must bind to 0.0.0.0: powershell uvicorn app.main:app --host 0.0.0.0 --port 8001
2. **Frontend** must bind to 0.0.0.0: powershell npm run dev -- --host 0.0.0.0
3. **Windows Firewall** must allow inbound on 8001 + 5173/5174 (run once as Admin): powershell New-NetFirewallRule -DisplayName "BVC24 Backend" -Direction Inbound -LocalPort 8001 -Protocol TCP -Action Allow New-NetFirewallRule -DisplayName "BVC24 Frontend" -Direction Inbound -LocalPort 5173,5174 -Protocol TCP -Action Allow
4. **Phone connects** to http://<PC-LAN-IP>:5174 (e.g. http://192.168.1.56:5174)

5. HTTPS for GPS: Mobile browsers require a **secure context** (HTTPS) for navigator.geolocation on most devices. For LAN testing, Chrome accepts localhost over HTTP, but a phone hitting http://192.168.1.56:5174 may have GPS blocked. Two options: - **Add an HTTPS reverse proxy** (Caddy, nginx with self-signed cert) - recommended for any production use. - **For trusted LAN dev only:** open chrome://flags/#unsafely-treat-insecure-origin-as-secure on the phone and add the LAN URL.

7.6 Daily Cron - Mark Absent

A scheduled job should run at end-of-day (e.g. 23:55) to:

```
For each ACTIVE employee with no Attendance row today:
  Insert Attendance(EMPLOYEE_ID, DATE, STATUS=ABSENT)
```

This is not yet automated by default - see §16.7 for the deployment task.

8. Leave & Permissions

8.1 Leave Types

Type	Quota-backed	Half-day allowed	Gender-gated
CASUAL	[OK]	[OK]	-
SICK	[OK]	[OK]	-
EARNED	[OK]	[OK]	-
MATERNITY	[OK]	[X]	FEMALE only
UNPAID	[X]	[OK]	-
LOP (Loss of Pay)	[X]	[OK]	-
PERMISSION	[X]	n/a (hourly)	-

Defaults defined in leave.py QUOTA_BACKED_TYPES and per-employee balances in the leave_balance table.

8.2 Apply for Day-Leave

Three channels - same backend:

1. **Inline form** on EmployeeDashboard.jsx
2. **Standalone page** at /apply-leave (ApplyLeave.jsx)
3. **Chatbot** - multi-turn conversation, see §13.2

All hit POST /leave/apply:

```
{
  "EMPLOYEE_ID": "<uuid or code>",
  "LEAVE_TYPE": "CASUAL",
  "START_DATE": "2026-06-15",
  "END_DATE": "2026-06-16",
  "HALF_DAY": false,
  "REASON": "Family function",
  "VENDOR_ID": 1
}
```

Validations: - Quota check (for quota-backed types) - Overlap check (against existing PENDING / APPROVED) - START_DATE <= END_DATE - HALF_DAY only valid for single-date requests - MATERNITY

requires GENDER=FEMALE - Every leave needs a non-empty REASON

8.3 Manager Approval (Email-Driven)

1. /leave/apply creates LeaveRequest with STATUS=PENDING_APPROVAL
2. Backend generates APPROVAL_TOKEN (signed, single-use)
3. Backend sends email to APPROVER (department head, or fallback)
Subject: "Leave request from <Employee> - <Dates>"
Body: [Approve] [Reject] buttons (HTML links)
4. Approver clicks Approve link
 - > GET /leave/decide/{token}?action=approve
 - > Validates token, flips STATUS=APPROVED, stamps APPROVAL_RESOLVED_AT
 - > Deducts quota (if quota-backed)
 - > Sends decision email back to employee
 - > Pushes in-app notification
 - > Renders styled HTML success page

No approver login required - the emailed link is the credential. Tokens are single-use and expire on first click.

8.4 Hourly Permissions

Different endpoint, different shape:

```
POST /leave/apply-permission
{
  "EMPLOYEE_ID":      "<uuid>",
  "PERMISSION_DATE":  "2026-06-15",
  "DURATION_HOURS":   2.5,
  "PERMISSION_SUBTYPE": "SHORT_PERMISSION",
  "REASON":           "Doctor appointment"
}
```

Subtypes: SHORT_PERMISSION, HALF_DAY, LATE_COMING, EARLY_EXIT.

Permissions create a LeaveRequest with LEAVE_TYPE=PERMISSION, DAYS=0. No quota deduction. Manager approval flow identical.

8.5 Cancel

```
PATCH /leave/{leave_id}/cancel
{ "EMPLOYEE_ID": "<uuid or code>", "NOTES": "Plans changed" }
```

If status was APPROVED -> refunds the quota. Ownership is checked (employee can only cancel own leaves; admins can cancel any).

8.6 Balances & History

Endpoint	Returns
GET /leave/balance/{employee_id}?year=2026	{ CASUAL {total, used, remaining}, SICK, EARNED }
GET /leave/my-requests?employee_id=<id>	All leave requests (newest first)
GET /leave/my-permissions?employee_id=<id>	PERMISSION rows only
GET /leave/dashboard-summary/{employee_id}	KPI counts for portal cards

9. Tasks & Productivity

9.1 Assignment Flow

```

HR / Manager  POST /task-assignment
               { EMPLOYEE_ID, PROJECT_ID, TASK_NAME, DUE_DATE, AUTO_ASSIGN }
               - if AUTO_ASSIGN=true -> load-balancer picks best available employee
               - Creates TaskAssignment with APPROVAL_STATUS=PENDING
               - Emails assignee
               - Pushes notification

Employee      Sees task in "Pending Acceptance" section
               PATCH /task-assignment/{id}/accept -> APPROVAL_STATUS=APPROVED
               PATCH /task-assignment/{id}/reject -> APPROVAL_STATUS=REJECTED
               -> may auto-reassign to next-best

```

9.2 Status Transitions

```

PENDING  --(Start Task)-->  IN_PROGRESS  --(Complete)-->  COMPLETED
                                   |
                                   L--(Hold)-->  ON_HOLD  --(Resume)-->  IN_PROGRESS

```

Endpoint:

```

PATCH /employee/{employee_id}/tasks/{assignment_id}/status
{ "status": "IN_PROGRESS" }

```

Side effects: - Stamps `START_TIME` (on `IN_PROGRESS`), `END_TIME` (on `COMPLETED`) - Awards points (configurable per-task or default) - Updates `WorkOrderStageProgress` if task is stage-linked - Re-computes employee performance score - Returns { `message`, `new_status`, `points_awarded`, `on_time`, `unlock_result` }

9.3 Points & Rewards

Points awarded per task completion: - Base: 10 points - On-time bonus: +5 - Streak multiplier: x1.0 -> x1.5 (resets on missed day)

Badges: - **Bronze** >= 100 pts - **Silver** >= 500 pts - **Gold** >= 1500 pts - **Platinum** >= 5000 pts

(Thresholds defined in `performance_service.py`; tune for your org.)

10. Memos & Acknowledgement

10.1 What Memos Are For

Memos are HR's structured audit trail for behavioural events:

Type	Severity	Example
APPRECIATION	INFO	"Outstanding work on Q2 deliverables"
WARNING	MEDIUM	"Late attendance 5 days this month"
DISCIPLINARY	HIGH	"Insubordination on shop floor"
INFORMATION	INFO	"Policy update - please read"
TRAINING	LOW	"Mandatory safety refresher"

10.2 Issuing a Memo (HR)

```

HR opens      EmployeeMemos page
HR clicks     "+ New Memo"
Fills         Employee, Type, Severity, Subject, Description,
              Issued By, Date, Attachment (optional)
Submits       POST /memos (multipart if attachment)
Backend       - Generates MEMO_NUMBER (MEMO-YYYY-nnnn)
              - Saves attachment to /static/memo-attachments/
              - Creates EmployeeMemo row
              - Sends in-app notification to employee

```

10.3 Employee Acknowledgement

Employee sees memos on dashboard under "My Memos". Each unacknowledged memo shows:

```

[!] WARNING #MEMO-2026-0042 [Acknowledge]
Subject:      Late attendance - 5 occurrences
Severity:     MEDIUM
Issued by:    Hari Krishnan (HR Manager)
Date:         2026-06-04

```

Click **Acknowledge**: - Modal asks for optional remark - POST /memos/{id}/acknowledge flips ACKNOWLEDGED_BY_EMPLOYEE=1, stamps ACKNOWLEDGED_DATE - HR sees the ack timestamp + remark in their view

10.4 Other Operations (HR only)

Action	Endpoint	Effect
Edit	PATCH /memos/{id}	Update subject/description/severity
Close	POST /memos/{id}/close	STATUS -> CLOSED
Cancel	POST /memos/{id}/cancel	STATUS -> CANCELLED
Delete	DELETE /memos/{id}	Soft-delete (DELETED_AT timestamp)
Export	GET /memos/export/csv?filters	CSV download for filtered set

10.5 Stats

GET /memos/stats?employee_id=<id> returns counts: total, active, closed, warnings, appreciations, disciplinary_open, pending_acknowledgement, last_memo_date, active_warnings.

11. Employee Documents

11.1 Document Types

```

AADHAAR    PAN        RESUME    OFFER_LETTER
CERTIFICATE EDUCATION  EXPERIENCE BANK_PROOF
ID_PROOF   OTHER

```

11.2 Upload (HR only)

```
POST /employees/{employee_id}/documents
multipart:
  file:          <binary>
  doc_type:      "AADHAAR"
  title:         "Aadhaar Card Front"
  notes:         "Issued 2018"
  uploaded_by_id: <admin uuid>
```

Backend saves to `/static/employee-docs/{employee_id}/{filename}` and creates an `EmployeeDocument` row referencing the URL.

11.3 View (Employee can see own)

```
GET /employees/{employee_id}/documents
GET /employees/{employee_id}/documents?doc_type=AADHAAR
```

Returns metadata + signed file URLs. Frontend shows them as a download list on the profile section.

11.4 Delete (HR only)

```
DELETE /employees/{employee_id}/documents/{doc_id}
```

Removes file from disk (best-effort) and deletes the row.

12. Performance, Productivity & Rewards

12.1 Scoring Inputs

`backend/app/services/performance_service.py` aggregates over a rolling window (default 30 days):

Input	Weight	Source
On-time completion rate	30%	<code>task_assignment.END_TIME</code> vs <code>DUE_DATE</code>
Attendance %	20%	<code>attendance.STATUS</code> count
Stage-flow contribution	15%	<code>work_order_stage_progress</code>
Quality (no NCRs)	15%	ncr rows where assignee = employee
Memo impact	10%	-ve for active WARNING / DISCIPLINARY
Workload throughput	10%	Tasks completed / period

Output: `performance_score` (0 - 100), `band` (*1 - *5), `suggested_increment_pct`.

12.2 Where the Score Surfaces

- Employee Dashboard top card (* rating)
- HR Dashboard performance row
- Chatbot: "performance summary" -> top + bottom + averages
- Payroll suggestions for annual increment

12.3 Refresh Cadence

The `portal-dashboard` endpoint returns a cached score (updated on task completion). For a lighter refresh:

```
GET /employee/{employee_id}/performance-only
```

Hit every 60 seconds from the dashboard.

13. The Floating ERP Assistant (Chatbot)

The chatbot is the **only feature that bridges every module** - employees can do nearly everything through conversation.

13.1 Architecture

```
Frontend  ChatBot.jsx
          - employeeId read from localStorage
          - leave_state persisted per-employee in localStorage
          - POST /chat/stream (Server-Sent Events)

Backend   chatbot.py
          chat_stream() generator yields SSE events:
            { type: "source", source: "leave" | "rules" }
            { type: "text", text: "..."}
            { type: "items", items: [...] }
            { type: "suggestions", suggestions: ["chip1", ...] }
            { type: "leave_state", state: {...}, ready_to_submit: bool }
            { type: "done" }

          Routing priority:
          1. Leave Bridge      (if employee_id set AND leave intent)
          2. Entity Lookup     (type any name -> profile)
          3. Rule-based Intent (~50 intents, synonym-aware)
          4. Topic Fallback    (friendly hint with chips)
```

13.2 Leave Multi-Turn Workflow

One-shot (everything in one message):

```
You: i need casual leave tomorrow for family function
Bot: Ready to submit:
     - Type:  CASUAL LEAVE
     - Date:  2026-06-07 (1 day)
     - Reason: family function
Your CASUAL balance: 12d -> 11d after this leave.
[Confirm & Submit] [Change date] [Change reason] [Cancel]

You: Confirm & Submit
Bot: [OK] Submitted! Your leave request #24 is now with HR for approval.
     [EMAIL] An email has been sent to the approver.
     [LIST] You can ask me 'leave status' anytime to check approval.
```

Multi-turn (bot fills missing slots):

```
You: i need leave next monday
Bot: Got: date 2026-06-08 - reason "i need leave next monday".
     What type of leave do you need?
     [Casual] [Sick] [Earned] [Unpaid]

You: sick leave, viral fever
Bot: Ready to submit: ...
     [Confirm & Submit] ...
```

13.3 Status Check

```
You: what is my leave status (or 'leave history', 'is my leave approved')
Bot: Here are your recent leave requests:
[~] #18 SICK - 2026-06-15 - Pending Approval
[OK] #16 CASUAL - 2026-05-22 - Approved
[X] #14 EARNED - 2026-04-10 - Rejected
Rejection: Project deadline this week
```

13.4 Other Things Both Employees & HR Can Ask

See the dedicated chatbot capability sheet for the full ~50-intent catalogue. Quick examples:

Category	Example questions
Greeting / help	hi, help, what can you do
Self leave	apply casual leave tomorrow, half day on 10th, cancel my leave
Leave status	leave status, my leave balance, leave history
HR overview	pending leave requests, who is on leave today, leave summary
Attendance	who is in office, late today, absent today, attendance today
Production	production status, work orders, machine models, show BOM
Quality	open NCRs, quality status
Suppliers	list suppliers, suppliers in Electronics
Performance	performance summary, top employees
Inventory	low stock, out of stock, inventory value
Tasks	pending tasks, overdue, workload summary
Entity lookup	Type any name: Hemnath, Snack Combo, ABC Foods

14. Employee API Reference

*Endpoint base: `http://<backend-host>:8001` All listed endpoints are employee-relevant.
Admin-only endpoints are in `docs/api/03-hr.md`.*

14.1 Authentication

Method	Path	Purpose
POST	<code>/employee-login</code>	Login + auto check-in
POST	<code>/employee-logout</code>	Logout + check-out
POST	<code>/employee-onboarding/{token}/login</code>	Token-gated public flow

14.2 Profile & Documents

Method	Path	Purpose
GET	/employees/by-code/{code}	Self-lookup
POST	/employees/by-code/{code}/submit-profile	One-shot self-reg
POST	/employees/{id}/upload-photo	Profile picture (multipart)
GET	/employees/{id}/documents	List own documents
GET	/employees/{id}/documents/{doc_id}	One document

14.3 Dashboard

Method	Path	Purpose
GET	/employee/{id}/portal-dashboard	One-shot dashboard load
GET	/employee/{id}/performance-only	Cheap performance refresh
GET	/connect/employee/{id}/360	Full 360° snapshot

14.4 Tasks

Method	Path	Purpose
GET	/employee/{ref}/today-task	Today's active task
GET	/employee/{ref}/tasks	All tasks grouped
GET	/employee/{ref}/pending-from-yesterday	Carryover
GET	/employee/{ref}/pending-acceptance	Awaiting accept/reject
PATCH	/task-assignment/{id}/accept	Accept task
PATCH	/task-assignment/{id}/reject	Reject task
PATCH	/employee/{id}/tasks/{aid}/status	Update status

14.5 Attendance

Method	Path	Purpose
POST	/check-in	Manual check-in (with GPS)
POST	/check-out	Manual check-out
GET	/attendance/today	Today's board
GET	/attendance/live-board	Live floor display (10s poll)

14.6 Leave

Method	Path	Purpose
POST	/leave/apply	Day-leave application
POST	/leave/apply-permission	Hourly permission
GET	/leave/balance/{id}	Quota check
GET	/leave/my-requests	Leave history
GET	/leave/my-permissions	Permission history
GET	/leave/dashboard-summary/{id}	KPI counts
PATCH	/leave/{id}/cancel	Cancel pending/approved
GET	/leave/decide/{token}	Approver action (HTML page)

14.7 Memos

Method	Path	Purpose
GET	/memos/employee/{id}	Own memos
POST	/memos/{id}/acknowledge	Acknowledge receipt

14.8 Geofence

Method	Path	Purpose
GET	/geofence/settings	Office centre + radius
POST	/geofence/validate	lat/lng -> allowed bool
POST	/geofence/log-failure	Audit failed attempt

14.9 Chatbot

Method	Path	Purpose
POST	/chat/stream	SSE multi-turn chat
GET	/chat/suggestions	Quick-reply chips
GET	/chat/health	Component status

14.10 Notifications

Method	Path	Purpose
GET	/notifications?employee_id=<id>	Inbox
GET	/notifications/unread-count?employee_id=<id>	Badge counter
PATCH	/notifications/{id}/read	Mark read

15. Notifications & Email

15.1 In-App Notifications

The `notification` table holds per-employee messages. Triggered by: - Task assignment - Leave approved / rejected - Memo issued - Production stage unlock - System alerts

The dashboard polls GET `/notifications/unread-count` every 30s and the bell icon shows the badge. Voice alerts fire if the employee has `localStorage.voice_enabled === 'true'`.

15.2 Email - SMTP Setup

Outbound mail is required for: - Leave approval links to managers - Decision emails back to employees - Task assignment notifications - Memo notifications (optional)

Configure in backend/.env:

```
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=your-sender@gmail.com
SMTP_PASS=your-app-password
SMTP_FROM="BVC24 ERP <noreply@bvc24.com>"

# OR use Resend
RESEND_API_KEY=re_XXXXXXXXXXXXX
RESEND_FROM=onboarding@resend.dev

# Useful in dev - re-routes all outbound to one address
EMAIL_TESTING_OVERRIDE_TO=qa-inbox@yourdomain.com
```

15.3 Approver Address Resolution

`leave.py` picks the approver for a request in this order: 1. `Employee.MANAGER_ID` -> that employee's email 2. Department head (`Department.HEAD_EMPLOYEE_ID`) 3. Vendor's default approver 4. Hard-coded admin (fallback, sends a warning)

If step 4 is hit and no fallback is set, the request is still created with `STATUS=PENDING_APPROVAL` but no email is sent. HR can approve manually from the Leave Management page.

16. Production Deployment Checklist

Before letting real employees use the system, work through every box.

16.1 Environment Variables

backend/.env:

```
# Database
DATABASE_URL=mysql+pymysql://user:pass@localhost:3306/bvc24_prod

# JWT
JWT_SECRET=<random-64-char-string>
JWT_ALGORITHM=HS256
JWT_EXP_MINUTES=720

# SMTP / Email
SMTP_HOST=...
SMTP_PORT=587
SMTP_USER=...
SMTP_PASS=...
SMTP_FROM="BVC24 <noreply@yourco.com>"

# Public URL (used in emailed approval links)
PUBLIC_FRONTEND_URL=https://erp.yourco.com
PUBLIC_BACKEND_URL=https://api.yourco.com

# Vendor scope
DEFAULT_VENDOR_ID=1

# Optional
EMAIL_TESTING_OVERRIDE_TO=
GEMINI_API_KEY=          # leave empty for rule-based-only chatbot
```

frontend/.env:

```
VITE_API_URL=https://api.yourco.com
```

16.2 Database Setup

```
# 1. Create database
mysql -u root -p -e "CREATE DATABASE bvc24_prod CHARACTER SET utf8mb4;"

# 2. Run migrations (tables auto-create on first uvicorn start via
Base.metadata.create_all)
cd backend && ./venv/Scripts/python.exe -m uvicorn app.main:app --host 127.0.0.1 --port
8001

# 3. Seed reference data (departments, designations, roles)
./venv/Scripts/python.exe scripts/seed_reference_data.py

# 4. Configure geofence
mysql -u root -p bvc24_prod -e "
    INSERT INTO geofence_settings
        (VENDOR_ID, OFFICE_NAME, LATITUDE, LONGITUDE, RADIUS_METERS, IS_ACTIVE)
    VALUES
        (1, 'Bharath Vending HQ', 11.0168, 76.9558, 100, 1);
"

# 5. Configure leave quotas (or use the LeavePolicy admin UI)
./venv/Scripts/python.exe scripts/seed_default_leave_policy.py
```

16.3 First Admin Account

```
./venv/Scripts/python.exe scripts/create_admin.py \
--email admin@yourco.com \
--password <strong-password> \
--name "System Administrator"
```

16.4 Backend Process

Use a process supervisor (NSSM on Windows, systemd on Linux, Docker, or PM2). Example NSSM (Windows):

```
nssm install BVC24-Backend "d:\PUVI-DOC\Vendor-based Manufacturing
ERP\backend\venv\Scripts\python.exe"
nssm set BVC24-Backend AppParameters "-m uvicorn app.main:app --host 0.0.0.0 --port 8001
--workers 2"
nssm set BVC24-Backend AppDirectory "d:\PUVI-DOC\Vendor-based Manufacturing ERP\backend"
nssm set BVC24-Backend AppStdout "d:\PUVI-DOC\Vendor-based Manufacturing
ERP\backend\uvicorn.log"
nssm set BVC24-Backend AppStderr "d:\PUVI-DOC\Vendor-based Manufacturing
ERP\backend\uvicorn.err"
nssm start BVC24-Backend
```

16.5 Frontend Build & Serve

```
cd frontend
npm install
npm run build          # generates dist/
# Serve with nginx, Caddy, or `serve -s dist -l 5173`
```

16.6 HTTPS Reverse Proxy

Required for mobile GPS. Minimal Caddy config (Caddyfile):

```
erp.yourco.com {
    root * /var/www/bvc24/frontend/dist
    try_files {path} /index.html
    file_server
}

api.yourco.com {
    reverse_proxy localhost:8001
}
```

Caddy auto-provisions Let's Encrypt certs. With this in place, mobile GPS works without flag overrides.

16.7 Cron / Scheduled Jobs

Required for daily housekeeping:

Job	Schedule	Action
Mark Absent	23:55 daily	Insert ABSENT rows for missing employees
Daily Backup	02:00 daily	mysqldump -> S3 or local backup folder
Performance Refresh	Hourly	Re-compute scoring (scripts/refresh_performance.py)
Leave Balance Accrual	Monthly (1st 00:05)	Top up CASUAL / SICK per policy
Stale Token Cleanup	Weekly	Invalidate unused leave-approval tokens older than 30 days

Sample Windows Task Scheduler entry for "Mark Absent":

```
schtasks /Create /TN "BVC24-MarkAbsent" /TR "d:\...\backend\venv\Scripts\python.exe
d:\...\backend\scripts\mark_absent.py" /SC DAILY /ST 23:55
```

16.8 Firewall & Network

Port	Purpose	Exposure
443	HTTPS (frontend + reverse-proxied API)	Public
80	HTTP -> HTTPS redirect	Public
8001	Backend uvicorn	LAN only (firewall block external)
3306	MySQL	Localhost only

16.9 Smoke Test Before Launch

Run these to validate end-to-end:

1. Create a test employee
2. Log in as that employee at the public URL
3. Complete self-registration form
4. Verify dashboard renders with no console errors
5. Check in from inside the office (geofence valid)
6. Apply for leave via chatbot -> confirm submit
7. Verify approval email arrived
8. Click approve link -> verify HTML page renders
9. Verify in-app notification arrived for employee
10. Ask chatbot: "what is my leave status" -> confirm shows approved
11. Check out -> verify worked hours computed
12. Try check-in from outside geofence -> confirm blocked
13. Open from phone on LAN -> confirm everything still works
14. Issue a memo -> employee acknowledges -> HR sees timestamp

16.10 Rollback Plan

If a release breaks:

```
# Stop service
nssm stop BVC24-Backend

# Revert code
cd backend
git checkout <last-known-good-commit>

# Restart
nssm start BVC24-Backend
```

Database migrations should always be additive (add columns, don't drop) so the previous version's code keeps working.

17. Security Considerations

17.1 Current State (Honest Assessment)

Concern	Status	Notes
Passwords bcrypt-hashed	[OK]	<code>connect.py</code> uses <code>passlib</code>
JWT issued on login	[OK]	12h expiry by default
JWT verified on endpoints	[!] Partial	Many endpoints are <code>Auth: None</code> - relying on frontend gating
HTTPS in production	[!] Required	Not enforced in code; deploy behind reverse proxy
GPS spoofing detection	[X]	Trusts browser-reported lat/lng
Approval-link tokens	[OK]	Single-use, signed
SQL injection	[OK]	SQLAlchemy parameterised queries
XSS in chatbot output	[!] Partial	React's default escaping protects; <code>dangerouslySetInnerHTML</code> not used
CSRF	[X]	No CSRF tokens; rely on SameSite cookies (set by reverse proxy)
File upload validation	[!] Partial	Extension checked; magic-byte check would harden

17.2 Hardening Before Public Internet Exposure

1. **Add JWT verification to every employee-only endpoint** - wrap with a `Depends(get_current_employee)` that decodes and validates.
2. **Rate-limit** `/employee-login` - 5 attempts per IP per 15 min.
3. **Force HTTPS** - backend sets `Strict-Transport-Security` header; frontend rejects `http://` API base.
4. **Rotate JWT secret** - generate fresh on first deploy; never commit `.env`.
5. **Audit file uploads** - Set max size (10MB photo, 25MB doc); verify magic bytes; quarantine to a non-served folder during scan.
6. **GPS sanity checks** - Reject lat/lng outside India bounding box; flag impossible velocity (>200 km/h between scans).
7. **Failed-login monitoring** - Email security@ on N+ failures from one IP.
8. **DB user least privilege** - App DB user has DML on `bvc24_prod` only; no DROP / CREATE / GRANT.

17.3 Data Retention

Data	Retention
Biometric scans	1 year
Attendance	Permanent (audit)
Leave requests	Permanent
Memos (active + closed)	7 years
Memos (deleted)	1 year (soft-deleted)
Security log	90 days
Notifications	90 days

Schedule a quarterly job to apply these retention windows.

18. Troubleshooting Runbook

18.1 Employee can't log in

1. Verify EMPLOYEE_CODE exists in the employee table:

```
SELECT ID, EMPLOYEE_CODE, STATUS FROM employee WHERE EMPLOYEE_CODE='EMP101';
```
2. Check STATUS:
 - ACTIVE -> fine
 - SUSPENDED -> "Account suspended" error
 - RESIGNED -> "Account inactive" error
 - TERMINATED -> same
3. Reset password if forgotten:

```
PUT /employees/{employee_id}/reset-password { NEW_PASSWORD: "..."} 
```
4. Check uvicorn logs for the actual error:

```
tail -f backend/uvicorn.err
```

18.2 GPS shows "Permission Denied" on phone

1. Browser settings -> Site settings -> Location -> Allow
2. iOS: Settings -> Safari -> Location -> Allow (also enable for Chrome)
3. If on http://<LAN-IP>:5174:
 Chrome flags -> "Insecure origins treated as secure" -> add URL
 OR deploy behind HTTPS (Caddy, see §16.6)
4. Verify with: navigator.geolocation in browser console
 - undefined -> not exposed (probably http issue)
 - object -> API present, check permission

18.3 Check-in says "Outside Office Geofence" but employee is inside

1. Verify geofence settings:
GET /geofence/settings
-> check LATITUDE, LONGITUDE, RADIUS_METERS
2. Get device's reported coords:
In browser console: navigator.geolocation.getCurrentPosition(p => console.log(p.coords));
3. Compute haversine distance manually.
If > RADIUS_METERS:
- Increase RADIUS_METERS (try 150-200m for noisy GPS)
- Or update OFFICE_LATITUDE/LONGITUDE to match actual centre
4. Indoor GPS is unreliable. Recommend kiosk-based biometric check-in for employees whose desks are deep inside a building.

18.4 Leave approval email not arriving

1. Check SMTP config in backend/.env
2. Check uvicorn.log for "Email sent" or error trace
3. Verify EMAIL_TESTING_OVERRIDE_TO isn't set (would re-route to QA)
4. Check approver resolution:
SELECT MANAGER_ID FROM employee WHERE ID=<applicant>;
SELECT EMAIL FROM employee WHERE ID=<manager>;
5. Gmail: enable App Password; SMTP_PASS isn't your account password
6. Resend: check API key + verified domain

18.5 Chatbot returns "source: rules" for leave request

This used to happen when the backend hadn't reloaded after a code change.

1. Verify only one uvicorn listens on 8001:
netstat -ano | findstr :8001 | findstr LISTENING
2. If multiple, stop all python processes and restart cleanly
3. Verify the chat health endpoint:
GET /chat/health
4. Check Logs for "Leave bridge" being hit

18.6 Dashboard shows stale data

1. portal-dashboard caches for 60s. Hard refresh: Ctrl+Shift+R
2. Check API call in browser DevTools -> Network - verify 200
3. Verify token isn't expired (would 401 and bounce to /login)
4. Backend log shows any SQLAlchemy errors

18.7 Task status update fails

1. Endpoint requires JWT in production (Authorization: Bearer <token>)
2. Check task ownership: employee can only update own tasks
3. Verify APPROVAL_STATUS=APPROVED (rejected tasks can't be progressed)
4. Status must be one of: PENDING, IN_PROGRESS, COMPLETED, ON_HOLD

18.8 Memo acknowledgement doesn't update HR view

1. Verify the POST succeeded (200 response with acknowledged_date)
2. HR view caches employee filter - change filter and back
3. SQL check:
SELECT ID, ACKNOWLEDGED_BY_EMPLOYEE, ACKNOWLEDGED_DATE
FROM employee_memo WHERE ID=<memo_id>;

19. HR / Admin Operational Tasks

19.1 Onboard a New Employee

1. Dashboard -> Employees -> "+ Add Employee"
2. Fill required fields. Set initial password.
3. Share EMPLOYEE_CODE + password with employee (secure channel)
4. Optionally upload photo + documents now
5. Employee will self-complete profile on first login

19.2 Reset Forgotten Password

1. Dashboard -> Employees -> click employee row
2. "Reset Password" -> enter new value
3. Share with employee

19.3 Approve a Leave Request

Email channel: click [Approve] in inbox -> done
 In-app channel: Dashboard -> Leave Management -> row -> Approve/Reject

19.4 Issue a Memo

Dashboard -> Memos -> "+ New Memo"
 Select employee, type, severity, fill subject + description,
 attach file if needed, submit. Employee gets notification.

19.5 Update Geofence

Dashboard -> Settings -> Geofence
 Update OFFICE_NAME, LATITUDE, LONGITUDE, RADIUS_METERS
 Toggle IS_ACTIVE to enable/disable enforcement

19.6 Export Memos to CSV

Memos page -> set filters -> "Export CSV"
 Or: GET /memos/export/csv?employee_id=...&memo_type=WARNING

19.7 Adjust Leave Policy

Dashboard -> Leave -> Policies
 Edit CASUAL_DAYS, SICK_DAYS, EARNED_DAYS, MATERNITY_DAYS
 Scope can be: GLOBAL, DEPARTMENT, DESIGNATION, INDIVIDUAL
 More specific scope wins. See backend/app/routes/leave.py for resolution rules.

19.8 Mark Employee Resigned / Terminated

Dashboard -> Employees -> row -> Edit
 Change STATUS to RESIGNED or TERMINATED
 Record DATE_OF_LEAVING
 Employee can no longer log in; data preserved for audit.

19.9 Bulk Mark Absent (if cron didn't run)

```
POST /attendance/mark-absent
{ "EMPLOYEE_ID": "<uuid>", "VENDOR_ID": 1, "NOTE": "No-show" }
```

Or run the script manually:

```
./venv/Scripts/python.exe scripts/mark_absent.py --date 2026-06-05
```

20. Extending the Module

20.1 Add a New Leave Type

1. backend/app/routes/leave.py -> add to VALID_LEAVE_TYPES
2. If quota-backed, add to QUOTA_BACKED_TYPES + LeaveBalance table column
3. backend/app/services/leave_chatbot_service.py -> add to VALID_LEAVE_TYPES for NLU
4. Frontend ApplyLeave.jsx -> add to the dropdown options
5. Update this doc's §8.1 table

20.2 Add a New Dashboard Section

1. Backend: add a new field to portal-dashboard response (don't break existing keys)
2. Frontend: read dashboardData.<new_field> in EmployeeDashboard.jsx
3. Render a new section component

20.3 Add a New Chatbot Intent

backend/app/routes/chatbot.py:

```
def handle_my_new_intent(tok_set, raw, db):
    # ... query DB ...
    return reply("My new answer", suggestions=["...", "..."])

# In build_rules(), add:
add(lambda t: has(t, "my_concept"), handle_my_new_intent, "my_intent")
```

That's it - the SSE streaming pipeline handles it.

20.4 Wire a New Notification Trigger

```
from app.models.models import Notification

db.add(Notification(
    EMPLOYEE_ID = emp.ID,
    TITLE       = "Something happened",
    BODY        = "Details here",
    TYPE        = "INFO",           # or WARNING, SUCCESS
    LINK        = "/some-path" # optional deep link
))
db.commit()
```

The dashboard's poll picks it up within 30s.

Appendix A - localStorage Keys

Key	Set by	Cleared by	Purpose
auth	Login	Logout, 401	Gate for ProtectedRoute
role	Login	Logout	"admin" or "employee"
token	Login	Logout, 401	JWT bearer
employee_id	Employee login	Logout	UUID of current employee
employee_name	Employee login	Logout	Display name
department	Employee login	Logout	For profile card
employee_role	Employee login	Logout	Designation / role label
loginTime	Employee login	Logout	Session age
attendance_status	Employee login	Logout	PRESENT / LATE flag
pending_yesterday	Employee login	Acknowledge	Carry-over banner trigger
pending_onboarding_token	sessionStorage; OnboardingChat	onboarding done	Locks login to Employee tab
chatbot_leave_state_{employee_id}	ChatBot leave flow	Cancel / submit	Multi-turn state per employee
voice_enabled	Settings toggle	User	Voice alert preference

Appendix B - State Transitions

Attendance.STATUS

```
(none)
|
|--- on time check-in    -> PRESENT
|--- late check-in      -> LATE (+ auto LATE_COMING Permission)
L-- no scan + cron      -> ABSENT
```

LeaveRequest.STATUS

```
(create) --> PENDING_APPROVAL
|
|---(approve link)--> APPROVED --> (employee cancels)--> CANCELLED +
refund
|---(reject link) --> REJECTED
L--(employee cancels)--> CANCELLED
```

TaskAssignment.APPROVAL STATUS

```
PENDING --> APPROVED (employee accepts) --> usable
L-> REJECTED (employee rejects) --> system tries auto-reassign
```

TaskAssignment.TASK STATUS

```
PENDING --> IN_PROGRESS --> COMPLETED
          ^       v
          L ON_HOLD
```

EmployeeMemo.STATUS

```
ACTIVE --> CLOSED (manual close after acknowledgement)
| --> CANCELLED (issued by mistake)
L --> (soft-deleted: DELETED_AT set, hidden from default queries)
```

Appendix C - Quick Reference for Employees

Print this page and pin it near workstations.

Logging In

1. Open the ERP URL in any browser (Chrome, Firefox, Safari).
2. Click the **Employee** tab.
3. Enter your **Employee Code** (e.g. EMP101) and password.
4. First time? You'll see a profile form - fill it once and submit.

Daily Workflow

- **Morning:** Logging in auto-marks your attendance. If you're outside the office, GPS will block check-in.
- **Find your task:** "Today's Tasks" section at the top of your dashboard.
- **Start work:** Click **Start Task** -> status flips to *In Progress*.
- **Done:** Click **Complete** -> you earn points + the next task unlocks.
- **Need to step out briefly:** Click **Hold**.
- **End of day:** Click **Logout** (top-right) to check out and record your worked hours.

Applying for Leave (three ways)

1. **Inline form** - bottom of your dashboard
2. **Standalone page** - /apply-leave
3. **Chatbot (easiest)** - click the red bubble bottom-right, type: "I need casual leave tomorrow for family function" The bot will confirm, then click **Confirm & Submit**.

Checking Leave Status

Type in the chatbot: **"what is my leave status"** - you'll see the status of every recent request.

Permission (Hourly)

Use the **Apply for Permission** form for short absences (doctor visit, errand). Different from day-leave; doesn't deduct quota.

Acknowledging a Memo

Whenever HR issues you a memo, it appears under **My Memos**. Click **Acknowledge** and optionally add a remark.

If Something Doesn't Work

Symptom	Try
Can't log in	Check caps lock; ask HR to reset password
"Outside Office Geofence"	Make sure you're inside the office; check phone has GPS on
GPS denied on phone	Browser settings -> Site permissions -> Location -> Allow
Chatbot doesn't respond	Refresh page; check internet
Forgot password	Ask HR to reset (Dashboard -> Employees -> your row -> Reset Password)

Who to Contact

Issue	Contact
Login / password	HR
Wrong attendance	HR (with proof of presence)
Leave rejected unfairly	Your manager (then HR)
Memo dispute	HR (use the "Remark" field when acknowledging)
System bug	IT / your team's tech lead

Document version: 1.0 - Last updated: 2026-06-06 - Maintained by: BVC24 Platform Team